

## Multiple Choice (10 marks)

1. Approximately what fraction of the human genome encodes proteins?
  - (a) 50%
  - (b) 5%
  - (c) 60%
  - (d) 25%
  - (e) 90%
2. Except for identical twins, no two people are genetically identical. Which of the following statements most accurately describes the main source of variation in humans?
  - (a) The main source of variation in humans is genetic engineering.
  - (b) The main source of variation in humans is due to dietary factors.
  - (c) Shuffling of alleles is responsible for the greatest variation in humans.
  - (d) The main source of variation in humans is due to exposure to radiation.
  - (e) The main source of variation in humans is genetic drift.
3. The ability of the brain to detect differences in stimulus intensity is best explained by the fact that which of the following varies with the stimulus intensity?
  - (a) The number of synapses crossed
  - (b) The speed of the action potential
  - (c) The amplitude of the action potential
  - (d) The number of neurons activated
  - (e) The refractory period of the neuron
4. If  $X$  has a normal distribution with mean 9 and standard deviation 3, find  $P(5 < X < 11)$ .
  - (a) - .09176
  - (b) .21185
  - (c) .37748
  - (d) .55962
  - (e) .65361
5. The first point of entry of water at the roots of a monocot plant is through the cytoplasm of cells of the

- (a) parenchyma
- (b) phloem
- (c) endoderm
- (d) pericycle
- (e) cortex

## True / False (10 marks)

*Answer True or False*

6. True or False: Do clinical variables predict pathologic radiographs in the first episode of wheezing?
7. True or False: Are failures of anterior cruciate ligament reconstruction associated with steep posterior tibial slopes?
8. True or False: Is leptin involved in phagocytic NADPH oxidase overactivity in obesity?
9. True or False: Is motion perception deficit in schizophrenia a consequence of eye-tracking abnormality?
10. True or False: Are patients with Werlhof's disease at increased risk for bleeding complications when undergoing cardiac surgery?

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the theoretical implications of genetic variation in a species with four alleles on chromosome 1 and three alleles on chromosome 2. How many distinct genotypes can be formed, and what factors may influence this genetic diversity?
12. Discuss the various methods that ensure cross-pollination in plants, focusing on the role of human intervention versus natural processes. Analyze their implications for plant reproduction.

*End of Paper*